

Assignment: Netflix User Analytics

Dataset Description

Column	Description
UserID	Unique user identifier
Age	User age
Gender	Male/Female
SubscriptionType	Basic/Premium/VIP
WatchHoursPerWeek	Hours watched per week
DevicesUsed	Number of devices used
FavoriteGenre	Action, Comedy, Drama, Sci-Fi, etc.
AdClicks	Number of advertisement clicks
MonthlySpend	Monthly spending (₹)
SubscriptionRenewed	Yes/No

Part A: Dataset Understanding

Q1. Load the dataset and display the first five records.

Code:

```
import pandas as pd

df = pd.read_csv("Dataset 2.csv")

df.head()
```

Output:

User ID	Age	Gender	SubscriptionType	WatchHoursPerWeek	DevicesUsed	FavoriteGenre	AdClicks	MonthlySpend	SubscriptionRenewed
1001	22	Female	Basic	23	5	Comedy	13	353	No
1002	55	Male	Basic	9	4	Drama	14	317	Yes

User ID	Age	Gender	SubscriptionType	WatchHoursPerWeek	DevicesUsed	FavoriteGenre	AdClicks	MonthlySpend	SubscriptionRenewed
1003	49	Male	Basic	8	3	Comedy	16	309	No
1004	39	Female	Premium	19	5	Drama	45	833	Yes
1005	38	Female	Premium	23	5	Sci-Fi	24	804	Yes

Q2. Determine the number of rows and columns

Code:

```
df.shape
```

Output:

```
(750, 10)
```

Q3. Display all column names

Code:

```
df.columns
```

Output:

```
Index(['UserID',
      'Age',
      'Gender',
      'SubscriptionType',
      'WatchHoursPerWeek',
      'DevicesUsed',
      'FavoriteGenre',
      'AdClicks',
      'MonthlySpend',
      'SubscriptionRenewed'],
```

```
dtype='object')
```

Q4. Identify numerical and categorical features

Code:

```
numerical_features = df.select_dtypes(include=['int64', 'float64']).columns
```

```
categorical_features = df.select_dtypes(include=['object']).columns
```

```
print("Numerical Features:")
```

```
print(numerical_features)
```

```
print("\nCategorical Features:")
```

```
print(categorical_features)
```

Output:

Numerical Features

```
Index(['UserID',
```

```
      'Age',
```

```
      'WatchHoursPerWeek',
```

```
      'DevicesUsed',
```

```
      'AdClicks',
```

```
      'MonthlySpend'],
```

```
dtype='object')
```

Categorical Features

```
Index(['Gender',
```

```
      'SubscriptionType',
```

```
      'FavoriteGenre',
```

```
      'SubscriptionRenewed'],
```

```
dtype='object')
```

Q5. Check whether the dataset contains missing values

Code:

```
df.isnull().sum()
```

Output:

```
UserID      0
```

```
Age         0
```

```
Gender      0
```

```
SubscriptionType  0
```

```
WatchHoursPerWeek  0
```

```
DevicesUsed    0
```

```
FavoriteGenre  0
```

```
AdClicks      0
```

```
MonthlySpend  0
```

```
SubscriptionRenewed  0
```

```
dtype: int64
```

Total Missing Values

```
df.isnull().sum().sum()
```

Output:

```
0
```

Part B: Exploratory Data Analysis

Q6. Calculate the average age of users.

Code:

```
df['Age'].mean()
```

Output:

```
41.824
```

Q7. Determine the average watch hours per week

Code:

```
df['WatchHoursPerWeek'].mean()
```

Output:

14.236

Q8. Find the average monthly spending of users

Code

```
df['MonthlySpend'].mean()
```

Output

689.90533333333334

Q9. Count the number of users in each subscription category

Code

```
df['SubscriptionType'].value_counts()
```

Output

Basic 342

Premium 279

VIP 129

Tabular Form

Subscription Type Number of Users

Basic 342

Premium 279

VIP 129

Total 750

Q10. Determine the percentage of users who renewed their subscriptions

Code

```
renewed_percentage = (  
    df['SubscriptionRenewed'].eq('Yes').mean()  
)
```

```
) * 100
```

```
print(renewed_percentage)
```

Output

```
46.266666666666666
```

Rounded:

```
46.27%
```

Supporting Counts

```
df['SubscriptionRenewed'].value_counts()
```

Output:

```
No    403
```

```
Yes   347
```

Part C: Data Preparation

Q11. Convert categorical features into numerical form

```
from sklearn.preprocessing import LabelEncoder
```

```
df_encoded = df.copy()
```

```
le_gender = LabelEncoder()
```

```
le_sub = LabelEncoder()
```

```
le_genre = LabelEncoder()
```

```
le_renew = LabelEncoder()
```

```
df_encoded['Gender'] = le_gender.fit_transform(df_encoded['Gender'])
```

```
df_encoded['SubscriptionType'] = le_sub.fit_transform(df_encoded['SubscriptionType'])
```

```
df_encoded['FavoriteGenre'] = le_genre.fit_transform(df_encoded['FavoriteGenre'])
```

```
df_encoded['SubscriptionRenewed'] =  
le_renew.fit_transform(df_encoded['SubscriptionRenewed'])
```

```
df_encoded.head()
```

Output

	UserID	Age	Gender	SubscriptionType	WatchHoursPerWeek	DevicesUsed	FavoriteGenre
AdClicks	MonthlySpend	SubscriptionRenewed					
0	1001	22	0	0	23	5	1 13 353 0
1	1002	55	1	0	9	4	2 14 317 1
2	1003	49	1	0	8	3	1 16 309 0
3	1004	39	0	1	19	5	2 45 833 1
4	1005	38	0	1	23	5	4 24 804 1

Q12. Define the feature set (X) and target variable (y)

```
X = df_encoded.drop('SubscriptionRenewed', axis=1)
y = df_encoded['SubscriptionRenewed']
```

```
print(X.shape)
print(y.shape)
```

Output

```
(750, 9)
(750,)
```

Q13. Split the dataset into training and testing sets

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    random_state=42
)
```

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

Output

```
(600, 9)
(150, 9)
(600,)
(150,)
```

Part D: Decision Tree Classification

Q14. Train a Decision Tree model to predict whether a user will renew their subscription

```
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(random_state=42)

dt_model.fit(X_train, y_train)

y_pred = dt_model.predict(X_test)
```

Output

Decision Tree model trained successfully.

Q15. Evaluate the model using accuracy

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

Output

Accuracy: 0.5666666666666667

Q16. Generate and interpret the confusion matrix

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)

print(cm)
```

Output

```
[[50 32]
 [33 35]]
```

Actual / Predicted No Renewal (0) Renewal (1)

No Renewal (0)	50	32
----------------	----	----

Actual / Predicted No Renewal (0) Renewal (1)

Renewal (1)	33	35
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Part E: K-Nearest Neighbors (KNN)

Q17. Train a KNN classifier with K = 5

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
```

```
knn = KNeighborsClassifier(n_neighbors=5)
```

```
knn.fit(X_train, y_train)
```

```
y_pred_knn = knn.predict(X_test)
```

```
knn_accuracy = accuracy_score(y_test, y_pred_knn)
```

```
print("KNN Accuracy:", knn_accuracy)
```

Output

KNN Accuracy: 0.60

Q18. Compare the accuracy of KNN with the Decision Tree model

```
print("Decision Tree Accuracy:", accuracy)
```

```
print("KNN Accuracy:", knn_accuracy)
```

Output

Decision Tree Accuracy: 0.5666666666666667

KNN Accuracy: 0.60

Model	Accuracy
--------------	-----------------

Decision Tree	56.67%
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KNN (K=5)	60.00%
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Part F: Linear Regression

Q19. Train a Linear Regression model to predict monthly spending

```

from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

X_reg = df_encoded.drop('MonthlySpend', axis=1)
y_reg = df_encoded['MonthlySpend']

X_train_reg, X_test_reg, y_train_reg, y_test_reg = train_test_split(
    X_reg, y_reg,
    test_size=0.20,
    random_state=42
)

lr = LinearRegression()

lr.fit(X_train_reg, y_train_reg)

r2_score = lr.score(X_test_reg, y_test_reg)

print("R2 Score:", r2_score)

```

Output

R² Score: 0.9660274555245928

Q20. Predict the monthly spending for a new user

```

new_user = [[1001,22,0,0,23,5,1,13,0]]

predicted_spending = lr.predict(new_user)

print(predicted_spending)

```

Output

[391.38516281]

Business Reflection Questions

1. Which factors appear to influence subscription renewal the most?

- WatchHoursPerWeek
- SubscriptionType

- MonthlySpend
- DevicesUsed
- AdClicks

2. Why is subscription renewal a classification problem?

Because the target variable has two categories:

- Yes
- No

The model predicts a class label rather than a continuous value.

3. Why is monthly spending a regression problem?

Because MonthlySpend is a continuous numerical value that can take many different values.

4. Which algorithm performed better for renewal prediction?

Algorithm	Accuracy
Decision Tree	56.67%
KNN (K=5)	60.00%

KNN performed better than Decision Tree.

5. How could the platform use these predictions to improve customer retention?

- Identify users likely to cancel subscriptions.
- Offer personalized discounts and promotions.
- Recommend relevant content based on viewing habits.
- Encourage upgrades to higher subscription plans.
- Improve engagement for low-activity users through targeted campaigns.